Royaume du Maroc

Ministère de l'Aménagement du Territoire National,
de l'Urbanisme, de l'Habitat
et de la Politique de la Ville

المملكة المغربية

وزارة إعداد التراب الوطني والتعمير والإسكان وسياسة المدينة

# RAPPORT D'ANALYSE

## Statistical Analysis Report

Rapport généré automatiquement

**Objet :** Statistical Analysis Report

**Organisme :** Ministère de l'Habitat

**Date de génération :** 09/08/2026 16:40

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# INFORMATIONS DU RAPPORT

Cette page présente les principales informations relatives au rapport généré automatiquement par la Plateforme IA d'Analyse du Secteur Immobilier.

| Champ | Valeur |
|---|---|
| Titre du rapport | Statistical Analysis Report |
| Sous-titre | Rapport généré automatiquement |
| Auteur | Ministère de l'Habitat |
| Date de génération | 09/08/2026 16:40 |
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| Nombre de tableaux | 1 |
| Nombre de graphiques | 1 |
| Lignes SQL | 15 |
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| Source des données | SQL Server - Ministère de l'Habitat |
| Généré par | Plateforme IA d'Analyse du Secteur Immobilier |

Les informations ci-dessus sont générées automatiquement au moment de la création du rapport. Elles permettent d'assurer la traçabilité du document et de faciliter son exploitation dans un contexte institutionnel.

# 1. Objet de la demande

Fais un rapport sur les prix des matériaux

## Indicateurs clés

| Résultats | Minimum | Maximum |
|---|---|---|
| 1987 lignes | 0,76 | 10 679,04 |
| Nombre de lignes retournées par la requête | Valeur minimale de PrixMoyen | Valeur maximale de PrixMoyen |

| Moyenne | Médiane |
|---|---|
| 419,61 | 40,39 |
| Moyenne de PrixMoyen | Médiane de PrixMoyen |

# 2. Résumé et analyse

# Executive Summary This report analyzes the dataset on material prices across regions and materials. The key statistics show substantial dispersion in average prices (PrixMoyen) across records, with a very high standard deviation relative to the mean, indicating significant variability in material pricing. The temporal indicator (AnneeFin) suggests data points span from 2019 to 2022, while the starting year (AnneeDebut) is consistently 2005. Regional and material-level breakdowns are available in the dataset, but specific per-item summaries are not provided in the supplied statistics. The analysis is based solely on the provided SQL-derived results and statistical summaries. # Objective - Assess the distribution and central tendency of material prices (PrixMoyen) and related index values (IndiceMoyen) in the dataset. - Describe the temporal context and variability of the data. - Provide visualization and business recommendations to support material price monitoring and procurement planning. # Dataset Overview - Dimensions: 1,987

rows, 6 columns. - Key columns: - Matiere (Material) - Region (Region) - PrixMoyen (Average Price) - IndiceMoyen (Average Index) - AnneeDebut (Start Year) - AnneeFin (End Year) - Observations from statistics: - AnneeDebut is constant at 2005.0 across all records. - AnneeFin ranges from 2019.0 to 2022.0, with a mean around 2021.24. - PrixMoyen shows wide dispersion (min ~0.76, max ~10,679.04) with a mean of ~419.61 and a large standard deviation (~1,509.26). - IndiceMoyen is relatively concentrated (min ~80.67, max ~138.16) with a mean of ~109.56 and a smaller standard deviation (~6.77). # Methodology - The report is generated from SQL query results and performed statistical analysis on the supplied dataset. - Descriptive statistics (min, max, average, median, standard deviation, total) were used to characterize PrixMoyen, IndiceMoyen, AnneeDebut, and AnneeFin. - Temporal analysis is constrained by the uniform AnneeDebut and varying AnneeFin values; without per-year grouping, trend conclusions are limited to published statistics. # Statistical Analysis - PrixMoyen (Average Price) - Minimum: 0.76 - Maximum: 10,679.04 - Average: 419.61 - Median: 40.39 - Standard Deviation: 1,509.26 - Total: 367,576.38 - Interpretation: The distribution is highly right-skewed, given the large gap between the median and mean and the very large maximum. A small subset of records drive extreme price values, while the majority cluster near lower values. - IndiceMoyen (Average Index) - Minimum: 80.67 - Maximum: 138.16 - Average: 109.56 - Median: 109.07 - Standard Deviation: 6.77 - Total: 217,697.14 - Interpretation: Prices are indexed with relatively moderate dispersion around the central tendency, indicating more stability in the index values compared to PrixMoyen. - AnneeDebut (Start Year) - All records: 2005 - Interpretation: Temporal analysis at the start year is not feasible due to lack of variation. - AnneeFin (End Year) - Minimum: 2019 - Maximum: 2022 - Average: 2021.24 - Median: 2022 - Standard Deviation: 1.30 - Total: 4,016,208 - Interpretation: The end-year values span four consecutive years, with most observations ending in 2022 and a slight spread indicating multi-year data coverage. # Key Findings - Price dispersion is high: PrixMoyen exhibits a very large standard deviation relative to its mean, and the maximum value is several orders of magnitude higher than the median. This indicates substantial variability in material prices across records or materials. - Central tendency vs. outliers: The median PrixMoyen is 40.39 while the mean is 419.61, suggesting that many records have comparatively low prices, but a minority have very high prices that pull the average upward. - Price index stability: IndiceMoyen shows tighter dispersion (mean ≈ 109.56, std ≈ 6.77) than PrixMoyen, suggesting more consistent indexing across records. - Temporal constraints: AnneeDebut is constant (2005), while AnneeFin varies (2019–2022). The lack of start-year variation limits the ability to identify price trends over time from this dataset alone. # Recommended Visualizations - Map: Display PrixMoyen by Matiere and Region to identify geographic patterns in pricing (supported by the presence of Region as a geographic column). - Bar chart: Matiere (x-axis) vs PrixMoyen (y-axis to compare average prices across materials) to highlight material-level pricing differences. - Scatter plot: Matiere (x-axis) vs PrixMoyen (y-axis) to examine price distribution across materials, benefiting from the presence of multiple numeric fields. # Business Recommendations - Monitor outliers in PrixMoyen: Given the high dispersion and

extreme maximum, implement anomaly detection or outlier flagging for unusually high material price observations to prevent procurement inefficiencies. - Region- and material-level segmentation: While not detailed in the provided statistics, the dataset supports breakdowns by Region and Matiere. Consider constructing regional price dashboards to identify geographic clusters of high or low pricing. - Use IndiceMoyen for normalization: The relatively stable IndiceMoyen suggests it can be used to normalize or compare prices across materials and regions, facilitating budget planning and index-linked contracts. - Temporal data enrichment: To enable trend analysis, incorporate start-year variation or additional yearly observations. With current constraints, trend conclusions should be avoided. # Conclusion The dataset presents substantial variability in material prices (PrixMoyen) across materials and regions, with a small subset of records driving very high values. The price index (IndiceMoyen) shows more stability, while temporal information is limited by a constant start year and end-year values spanning 2019–2022. The findings support targeted monitoring of outliers and the development of region- and material-specific price views to inform procurement strategies. Visualization recommendations aim to illuminate geographic and material-level price patterns, reinforcing data-driven decision making.

# 3. Requête SQL exécutée

SELECT n.nom AS Matiere, r.nom_region AS Region, AVG(f.prix_moyen) AS PrixMoyen, AVG(f.indice) AS IndiceMoyen, MIN(f.annee) AS AnneeDebut, MAX(f.annee) AS AnneeFin FROM FACT_PRIX_INDICES f INNER JOIN REGION r ON f.id_region = r.id_region INNER JOIN NIVEAU_MATERIAU n ON f.id_niveau_mat = n.id GROUP BY n.nom, r.nom_region ORDER BY r.nom_region, n.nom

# 4. Méthodologie

Ce rapport est généré automatiquement à partir des données disponibles dans la base de données du Ministère. La question formulée en langage naturel est interprétée par l'agent d'intelligence artificielle, puis transformée en requête SQL. Les résultats retournés sont ensuite analysés afin de produire une synthèse, des indicateurs clés et des visualisations.

# 5. Conclusion

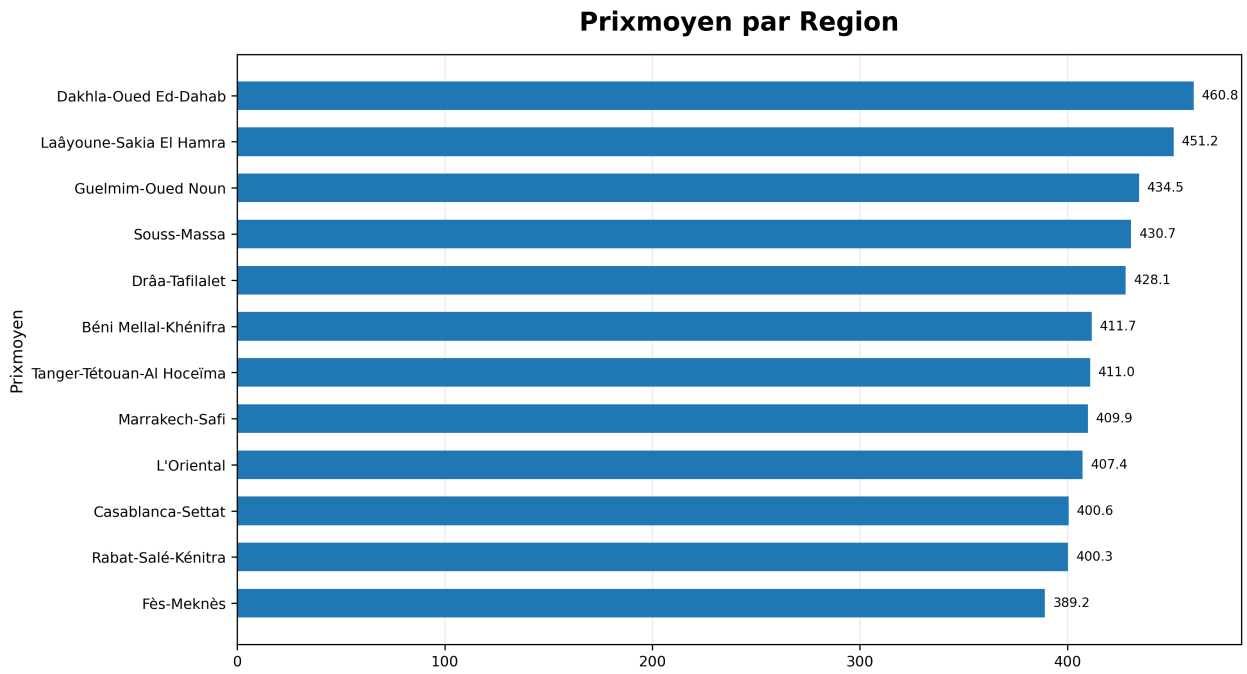
Les résultats présentés dans ce rapport permettent de synthétiser les données retournées par la requête et d'en faciliter l'interprétation à travers des indicateurs et des visualisations adaptés.

# 6. Aperçu des résultats (15 lignes sur 1987)

| Matiere | Region | PrixMoyen | IndiceMoyen | AnneeDebut | AnneeFin |
|---|---|---|---|---|---|
| 1 Compartiment (u) | Béni Mellal-Khénifra | 370,16 | 80,67 | 2005 | 2019 |
| 12-16/20 (ml) | Béni Mellal-Khénifra | 19,13 | 118,87 | 2005 | 2019 |
| 12-16/20/52 (u) | Béni Mellal-Khénifra | 4,89 | 123,12 | 2005 | 2019 |
| 15-20/20/40-50 (u) | Béni Mellal-Khénifra | 4,18 | 127,43 | 2005 | 2019 |
| 15x15 à 20x60 (m²) | Béni Mellal-Khénifra | 66,23 | 120,28 | 2005 | 2019 |
| 20/20 (ml) | Béni Mellal-Khénifra | 26,78 | 118,87 | 2005 | 2019 |
| 20/20/52 (u) | Béni Mellal-Khénifra | 6,32 | 123,12 | 2005 | 2019 |
| 20x20 (m²) | Béni Mellal-Khénifra | 40,89 | 106,94 | 2005 | 2019 |
| 27S (m²) | Béni Mellal-Khénifra | 18,19 | 113,53 | 2005 | 2019 |
| 2ème choix (kg) | Béni Mellal-Khénifra | 52,41 | 109,62 | 2005 | 2019 |
| 2ème Choix (m3) | Béni Mellal-Khénifra | 9 247,89 | 114,56 | 2005 | 2019 |
| 33x33 & similraires (m²) | Béni Mellal-Khénifra | 70,87 | 120,52 | 2005 | 2019 |
| 33x33 à 50x50 (m²) | Béni Mellal-Khénifra | 71,76 | 111,84 | 2005 | 2019 |
| 36S (m²) | Béni Mellal-Khénifra | 23,86 | 113,53 | 2005 | 2019 |
| 4 mm (m²) | Béni Mellal-Khénifra | 63,75 | 109,89 | 2005 | 2019 |

# 7. Visualisations

Les graphiques suivants ont été générés automatiquement afin de mettre en évidence les tendances, comparaisons et indicateurs pertinents issus des résultats SQL.




*Comparaison de Prixmoyen selon Region.*